

1. BP = CO × TPR

WHAT YOU SEE

Scales: cardiac output balanced against peripheral resistance, BP=CO×TPR banner

WHAT IT ENCODES

Mean arterial pressure = cardiac output × total peripheral resistance (MAP = CO × TPR). These are the two master levers every antihypertensive drug pulls: drugs lower CO (beta-blockers, non-DHP CCBs, diuretics via volume) and/or lower TPR (ACEi/ARB, DHP CCBs, alpha-blockers, vasodilators). MAP ≈ DBP + 1/3 pulse pressure. Essential HTN (~90-95% of cases) is driven chiefly by increased TPR.

BOARD TRAP

WRONG: assuming BP is set by heart rate alone, so any tachycardia means hypertension. Discriminator — BP depends on the PRODUCT of CO and TPR; a high HR with low stroke volume or low resistance need not raise BP, and chronic essential HTN is usually a TPR (resistance) problem, not a rate problem.

2. Cardiac output

WHAT YOU SEE

Left pan labeled CARDIAC OUTPUT, SV×HR heart below

WHAT IT ENCODES

CO = stroke volume × heart rate. Stroke volume rises with preload (venous return/volume) and contractility and falls with afterload. Volume expansion (excess sodium/aldosterone, renal failure) and sympathetic drive raise CO. Diuretics lower CO by reducing preload; beta-blockers lower it by reducing HR and contractility.

BOARD TRAP

WRONG: thinking volume-mediated (high-CO) HTN responds best to a vasodilator. Discriminator — when HTN is volume/sodium-driven (CKD, primary aldosteronism, salt sensitivity), a diuretic that reduces preload/CO is far more effective than simply vasodilating; match the drug to the dominant lever.

3. Peripheral resistance

WHAT YOU SEE

Right pan PERIPHERAL RESISTANCE, functional vs structural arrows

WHAT IT ENCODES

TPR is governed by arteriolar tone. It rises from FUNCTIONAL vasoconstriction (sympathetic alpha-1 tone, angiotensin II, endothelin, low nitric oxide) and from STRUCTURAL remodeling (arteriolar wall hypertrophy, arterial stiffening). Poiseuille's law: resistance is exquisitely sensitive to radius ($\propto 1/r^4$), so small caliber changes greatly affect BP. Most essential HTN is a high-TPR state.

BOARD TRAP

WRONG: expecting structural arterial stiffening to reverse quickly once a vasodilator is started. Discriminator — functional vasoconstriction reverses fast, but structural remodeling/stiffening responds slowly and incompletely; this underlies isolated systolic HTN in the elderly and the lag in BP control.

4. Na & kidney

WHAT YOU SEE

Faucet, Na⁺ excretion, kidney sets long-term BP

WHAT IT ENCODES

The kidney sets LONG-TERM BP through pressure natriuresis: when BP rises, the kidney excretes more sodium and water, dropping volume and BP back toward setpoint. Any defect that shifts this curve rightward (reduced nephron number, excess aldosterone, renal artery stenosis, high salt intake in salt-sensitive people) sustains hypertension. This is why sodium handling and the kidney are central to chronic BP.

BOARD TRAP

WRONG: believing a perfectly functioning kidney can sustain chronic hypertension without resetting its natriuresis curve. Discriminator — Guyton's principle holds that the kidney would normally correct any sustained BP rise; chronic HTN requires a shifted pressure-natriuresis relationship, so secondary renal/adrenal causes deserve consideration.

5. BP classification pyramid

WHAT YOU SEE

Color tiers: Normal <120/80, Elevated 120-129/<80, Stage1 130-139/80-89, Stage2 ≥140/≥90

WHAT IT ENCODES

ACC/AHA 2017 categories: Normal <120 AND <80; Elevated 120-129 AND <80; Stage 1 130-139 OR 80-89; Stage 2 ≥140 OR ≥90. Hypertensive crisis is >180 and/or >120 (urgency without, emergency with end-organ damage). Diagnosis requires the average of ≥2 readings on ≥2 separate occasions. General treatment target <130/80.

BOARD TRAP

WRONG: using the older JNC7 threshold of 140/90 to define hypertension. Discriminator — ACC/AHA lowered Stage 1 to 130-139/80-89, so a patient at 135/85 is hypertensive (Stage 1) by current US criteria, even though older guidelines called it 'prehypertension.'

6. Take higher category

WHAT YOU SEE

Banner: TAKE HIGHER CATEGORY above pyramid

WHAT IT ENCODES

When systolic and diastolic readings fall in different stages, classify by the HIGHER category. Example: 138/92 → SBP is Stage 1 but DBP is Stage 2, so the patient is Stage 2. This 'whichever is worse' rule ensures the more severe value drives staging and treatment intensity.

BOARD TRAP

WRONG: staging 138/92 as Stage 1 because the systolic falls in the 130-139 range. Discriminator — the diastolic of 92 is Stage 2 (≥90), and the rule is to take the higher category, so this patient is Stage 2 and warrants Stage 2 management.

7. Accurate BP technique

WHAT YOU SEE

Checklist: no caffeine, no smoke, no exercise, rest 5min, back support, feet flat, arm at heart, avg 2 readings

WHAT IT ENCODES

Proper measurement: no caffeine/smoking/exercise for 30 min before, empty bladder, rest seated 5 min, back supported, feet flat on floor (uncrossed legs), arm supported at HEART level, correct cuff, no talking, and average ≥2 readings on ≥2 visits. Errors inflate readings (e.g., unsupported back/arm, crossed legs, talking each add several mmHg).

BOARD TRAP

WRONG: diagnosing HTN and starting drugs from a single elevated reading taken with the patient just arrived, talking, arm dangling. Discriminator — technique errors and white-coat effect inflate a one-off reading; diagnosis requires correctly measured, averaged readings across multiple visits or out-of-office confirmation.

8. Correct cuff size

WHAT YOU SEE

WRONG CUFF crossed out, validated device, booth patient

WHAT IT ENCODES

Cuff bladder should encircle ~80% of arm circumference. A cuff that is too SMALL (or applied over a large arm) OVERestimates BP; a cuff too LARGE underestimates it. Always use an appropriately sized cuff on a validated/calibrated device, positioned at heart level.

BOARD TRAP

WRONG: diagnosing hypertension in an obese patient measured with a standard (too-small) cuff. Discriminator — an undersized cuff falsely RAISES the reading; before labeling such a patient hypertensive, re-measure with a large/thigh cuff sized to the arm.

9. Office BP limited

WHAT YOU SEE

The OFFICE BP sign with a '+7 mmHg' arrow next to the doctor — the inflated number pictures how office readings run falsely high.

WHAT IT ENCODES

Office BP is the least reliable setting — subject to white-coat effect, technique error, and single-time-point variability. Current guidance favors confirming a new diagnosis and assessing control with OUT-of-office measurement (HBPM or ABPM) before committing to lifelong therapy.

BOARD TRAP

WRONG: titrating drugs solely on office readings while ignoring discordant home values. Discriminator — office readings can over- (white coat) or under-estimate (masked) true BP; out-of-office monitoring should reconcile discordance before escalating or de-escalating therapy.

10. White-coat HTN

WHAT YOU SEE

WHITE COAT booth, doctor, high in clinic / home OK

WHAT IT ENCODES

White-coat hypertension = elevated OFFICE BP but NORMAL out-of-office (home/ambulatory) BP, due to alerting response in clinic. It carries lower (but not zero) CV risk than sustained HTN. Confirm with ABPM/HBPM before treating; manage with lifestyle and monitoring rather than reflexive drug therapy.

BOARD TRAP

WRONG: starting/intensifying antihypertensives based on high clinic readings in a patient whose home BP is normal. Discriminator — this is white-coat HTN; treating it risks overtreatment and hypotension. Confirm with out-of-office monitoring; observe and address lifestyle instead of adding drugs.

11. Masked HTN

WHAT YOU SEE

MASKED booth, normal office but HIGH home, RISK + HTN

WHAT IT ENCODES

Masked hypertension = NORMAL office BP but ELEVATED out-of-office BP. It is dangerous because it is easily missed yet carries cardiovascular risk similar to sustained HTN (target-organ damage, events). Suspect it with normal office readings but signs of end-organ damage (LVH, retinopathy, proteinuria) or high-risk profile; confirm with ABPM/HBPM and treat.

BOARD TRAP

WRONG: reassuring a patient with normal office BP but LVH/proteinuria that they don't have hypertension. Discriminator — those end-organ findings point to MASKED HTN; out-of-office monitoring will reveal elevated BP that the clinic missed, and treatment is warranted.

12. HBPM

WHAT YOU SEE

HBPM home monitor kiosk

WHAT IT ENCODES

Home BP monitoring (HBPM): patient self-measures with a validated upper-arm device, typically 2 readings AM and PM over ~7 days, discarding day 1, then averaging. Home HTN threshold is $\geq 130/80$ (lower than the office $\geq 140/90$ reference). HBPM confirms diagnosis, detects white-coat/masked patterns, improves adherence, and tracks control.

BOARD TRAP

WRONG: applying the office $\geq 140/90$ cutoff to home readings. Discriminator — out-of-office thresholds are LOWER (home/ABPM-daytime $\geq 130/80$); using the office number to interpret home values under-diagnoses true hypertension.

13. ABPM

WHAT YOU SEE

ABPM 24-hour ambulatory monitor

WHAT IT ENCODES

Ambulatory BP monitoring (ABPM) is the GOLD STANDARD to confirm the diagnosis: a cuff records BP every 15-30 min over 24 hours, giving daytime, nighttime, and 24-hour averages. It uniquely detects white-coat and masked HTN and nocturnal dipping. Thresholds: 24h average $\geq 125/75$, daytime $\geq 130/80$, nighttime $\geq 110/65$. Non-dipping (night BP not falling $\geq 10\%$) predicts worse CV outcomes.

BOARD TRAP

WRONG: ignoring a 'non-dipping' nocturnal pattern as clinically irrelevant. Discriminator — loss of the normal $\geq 10\%$ nighttime BP fall (non-dipping/reverse dipping) is an independent predictor of cardiovascular events and may signal secondary causes such as OSA, CKD, or autonomic dysfunction.

14. Age trends

WHAT YOU SEE

Graphs: SBP rises with age, DBP rises then falls, ISH in elderly

WHAT IT ENCODES

With aging, large arteries stiffen: SBP keeps RISING throughout life while DBP rises until ~50-60 then FALLS, widening the pulse pressure. The result is isolated systolic hypertension (ISH: SBP ≥ 130 with DBP < 80), the dominant HTN pattern in the elderly and a strong CV risk marker. Preferred agents in ISH: thiazides and DHP CCBs.

BOARD TRAP

WRONG: leaving isolated systolic HTN untreated in an elderly patient because the diastolic is normal/low. Discriminator — ISH is a treatable, high-risk condition (SPRINT-supported benefit of SBP lowering in older adults); a low DBP does not make a high SBP benign, though one avoids dropping DBP too far.

15. RAAS cascade

WHAT YOU SEE

The plumbing pipe running along the wall feeding the Na⁺/kidney faucet — the pipe IS the renin-angiotensin cascade delivering pressure to the kidney's sodium tap.

WHAT IT ENCODES

NEW: The renin-angiotensin-aldosterone system is the key humoral BP regulator. Low renal perfusion/Na delivery or sympathetic (beta-1) stimulation -> juxtaglomerular cells release RENIN -> angiotensinogen to angiotensin I -> ACE (lung) to angiotensin II. Angiotensin II constricts arterioles (raises TPR), constricts the efferent arteriole (maintains GFR), stimulates thirst/ADH, and triggers ALDOSTERONE (Na/water retention, K⁺ excretion). This cascade is the drug target of ACEi, ARBs, MRAs, beta-blockers, and renin inhibitors.

BOARD TRAP

WRONG: expecting renin to be HIGH in primary aldosteronism. Discriminator — in Conn syndrome aldosterone is autonomously high so renin is SUPPRESSED (high aldosterone:renin ratio); high renin instead points to secondary causes like renovascular disease, where reduced renal perfusion drives renin up.

16. Baroreceptor reflex

WHAT YOU SEE

The City Hall mayor figure standing on the balance, hands steadying the CO-vs-TPR scale — he is the autonomic/baroreflex controller making fast second-to-second BP adjustments.

WHAT IT ENCODES

NEW: Baroreceptors in the carotid sinus (CN IX) and aortic arch (CN X) provide rapid, SECOND-to-second BP control. A BP rise stretches them -> increased afferent firing -> medulla decreases sympathetic and increases vagal tone -> lower HR, contractility, and vasoconstriction, normalizing BP (and vice versa for a fall). This reflex buffers acute changes (e.g., standing) but RESETS to defend a higher BP in chronic HTN, so it does not correct sustained hypertension.

BOARD TRAP

WRONG: relying on the baroreflex to explain why chronic hypertension should self-correct. Discriminator — baroreceptors handle SHORT-term fluctuations and reset to a higher setpoint in chronic HTN; long-term BP is set by the kidney (pressure natriuresis), which is why the reflex cannot fix sustained hypertension.